
Assignment 2: Model Validation

QUANTITATIVE FINANCIAL RISK MANAGEMENT

Group 20 - Validating Group 9

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1 Documentation

Dataset provided was only raw data. Using documentation, their data cleaning and processing steps were recreated as faithfully as possible.

1.1 Data and Portfolio

The documentation for asset selection, missing data handling, and return calculation seem extensive. There do, however, remain some issues. There are dates where multiple assets had missing observations, usually falling on market closures and holidays. While they mention these for streaks of missing days, they do not mention days with multiple missing assets, which are automatically forward filled. The dataset is also of the years 2006–2013, and while it does include a crisis period, using recommendations from a previous market environment do not guarantee they hold for current market regimes.

The EURUSD data also has some discrepancies, there are are notable spikes on the 8th on numerous months in 2008. Looking at YahooFinance, it appears they have data issues on these dates. This, coupled with the long streaks of missing EURUSD data, indicate that an alternative data source might be needed.

Furthermore, EUR assets remain in EUR and USD assets in USD, and a base currency for the portfolio is not specified. FX risk is captured as a 10% weight of EURUSD in the portfolio, but the reasoning behind this is not documented. FX risks on EUR assets if investor is US-based or vice-versa are therefore missed. The loan is modelled as a floating rate instrument, but the time to reset is not specified. This also means their portfolio has practically no interest rate risk, if the loan is constantly reset. The loan component return always stays above zero, and hence the VaR and ES are also zero for most models.

The portfolio weights are well documented and assigned simply. The use of variances which are forward looking at time of portfolio construction, however, is visual and possibly problematic, as the low variance asset is given a lower weight, but this cannot possibly be known pre-2006. The weights are also mentioned to be fixed through the sample period, which signals rebalancing daily, but this could be documented more explicitly.

1.2 Models and Outcomes

The documentation for the model types are thorough, with complete documentation for the affect of sample periods. For EWMA, the smoothing parameter $\lambda = 0.94$ is given but not justified. The GARCH backtest uses a reduced grid of every fifth day, which is documented, but the affect of this on the results such as lower power are not well documented.

The stress testing results for currency rate shocks underestimate the actual impact as they only look at the changes on the currency component of the portfolio and neglect base currency of portfolio. Interest rate sensitivity is also mislabelled in table 20, with basis point shifts represented instead of percentage point shifts.

Overall the model, their results and recommendations are extensively documented, with minor discrepancies due to earlier data handling issues. The report is quite long, and could be presented more concisely. Some parts could be further justified, and other do not require as

much justification as is provided.

2 Theoretical Assessment

2.1 Methodology and assumptions

The overall framework is reasonable for the assignment and follows standard market risk practice. The group compares Normal VaR, Student-t VaR, Historical Simulation, GARCH, FHS and backtesting, which gives a useful mix of simple and more advanced methods. The main weaknesses are in the assumptions behind some portfolio components. In particular, FX risk is treated rather simply, since EUR assets remain in EUR and USD assets remain in USD, while EUR/USD is added as a separate portfolio position without a clearly defined base currency. The loan is also modelled in a simplified way as DTB3 plus a fixed spread, which leaves it with very limited interest rate risk.

2.2 Appropriateness

Overall, the methodology is appropriate for the purpose of the assignment. Using several risk models instead of relying on one single approach is a strength, and the inclusion of the 2008 crisis period makes the analysis more meaningful. However, some choices reduce the strength of the conclusions. The FX setup may not fully capture the actual currency exposure of the portfolio, and the loan specification is too stylised to reflect realistic interest rate sensitivity. These issues do not make the model unusable, but they do limit how realistic it is.

2.3 Modelling techniques

The modelling techniques are generally well documented. The report clearly uses Normal and Student-t Variance-Covariance VaR, Historical Simulation, GARCH and FHS. The simpler models are transparent and can be checked directly from the data. In particular, the Normal, Student-t and Historical Simulation results can be reproduced quite closely, which supports the correctness of those parts. The more advanced models are less easy to verify fully because not all implementation details are provided, so they are less transparent from a validation point of view.

2.4 Calibration

Calibration is documented, but not always justified in enough detail. The EWMA parameter $\lambda = 0.94$ is stated, but the reason for this choice is not discussed. The GARCH backtest is performed only every fifth day, which is documented, but the effect of this reduced grid on comparability and test power is not explained sufficiently. The portfolio weights are clearly stated, but if they are fixed through time this effectively implies rebalancing, which could have been made more explicit. So, while the calibration choices are not unreasonable, some of them are not fully motivated.

2.5 Model inputs

The documentation of asset selection, missing data handling and return calculation is fairly extensive, but some input issues remain. The report discusses long missing EUR/USD gaps, yet days with multiple missing assets are largely forward filled without much further comment. This is important because the sample includes stressed periods, where data treatment can materially affect risk estimates. The EUR/USD series also appears weaker than the other inputs, with long missing stretches and some irregular spikes in 2008, suggesting that cross-checking with another data source would have been preferable. More generally, the data cover 2006–2013, which is useful because it includes a crisis period, but conclusions based on that sample may not automatically carry over to different market conditions.

3 Quantitative Assessment

Implementation accuracy: Historical Simulation

We rebuilt the daily log-returns of each asset, used the same EUR 1,000,000 portfolio, and took the one-day 99% VaR as the empirical 1% quantile of the loss distribution. Component VaRs are computed asset by asset; the portfolio VaR is computed from the aggregated portfolio P&L, not as a sum of components. Our numbers and theirs are in Table 1.

Component	Our VaR	Their VaR	Δ
INGA.AS	11.3571%	11.3571%	0.0000%
SHEL	5.4735%	5.4735%	0.0000%
XOM	4.6385%	4.6385%	0.0000%
^AEX	4.4184%	4.4184%	0.0000%
^GSPC	4.4274%	4.4274%	0.0000%
EURUSD=X	1.8977%	1.8977%	0.0000%
log_loan	−0.0040%	0.0000%	−0.0040%
Portfolio	3.5105%	3.5105%	0.0000%

Table 1: Component and portfolio one-day 99% Historical VaR. “Our VaR” is from our re-implementation; “Their VaR” is from the peer group’s report.

Every equity, index and FX component matches to four decimals, and the portfolio VaR matches at 3.5105%. So our pipeline agrees with theirs: if we end up disagreeing on FHS–EWMA, the cause is in the volatility scaling, not in the data.

Implementation accuracy: FHS–EWMA

We re-implemented FHS–EWMA on the same cleaned data, with the same $\lambda = 0.94$ that the report uses. We standardise each daily return by its EWMA volatility, and we simulate next-day returns by rescaling the standardised innovations with the latest EWMA volatility forecast. The one-day 99% VaR is then the empirical 1% quantile of the scaled distribution. Our numbers and theirs are in Table 2.

The numbers are close, but they do not match. Our VaR is between 1.4% and 3.5% lower than theirs in relative terms across the equity, index and FX components, with EURUSD=X

Component	Our VaR	Their VaR	Δ
INGA.AS	4.0785%	4.1952%	−0.1167%
SHEL	2.4645%	2.5283%	−0.0638%
XOM	3.0775%	3.1377%	−0.0602%
^AEX	2.2553%	2.3274%	−0.0721%
^GSPC	1.6691%	1.7301%	−0.0610%
EURUSD=X	0.8462%	0.8345%	0.0117%
log_loan	−0.0030%	0.0000%	−0.0030%
Portfolio	1.4982%	1.5300%	−0.0318%

Table 2: Component and portfolio one-day 99% VaR under Filtered Historical Simulation with EWMA volatility scaling ($\lambda = 0.94$).

going the other way. At the portfolio level we get 1.4982% against their 1.5300%, a gap of about 2%. The loan still differs the same way it did under HS.

We could not close this gap, even though we used the same data, the same λ , and what we think is the same procedure. The size and direction of the differences look more like the kind of gap you get from small implementation choices that the report does not nail down, rather than from a real modelling disagreement. The most likely culprits are:

- how the EWMA recursion is started (sample variance over a warm-up window, the first squared return, or some unconditional estimate);
- which volatility is used to standardise day t 's return - the forecast made at $t - 1$, or the EWMA value updated through t ;
- whether returns are demeaned before they enter the variance recursion;
- which interpolation rule is used to read off the 1% empirical quantile.

Our reading is that the FHS–EWMA model is reproducible in spirit but not exactly replicable from the report alone. We flag this as a documentation issue: a model meant to feed into a risk decision should be written down precisely enough that an independent re-implementation lands on the same number, and right now the report does not give us enough to do that.

Implementation accuracy: stress testing

We re-ran the same single-factor stress scenarios on the cleaned data: equity shocks of $\pm 20\%$ and $\pm 40\%$, FX shocks of $\pm 10\%$ and $\pm 20\%$ on EUR/USD, and parallel rate shifts of ± 2 bp and ± 3 bp. For each scenario we compute the immediate mark-to-market P&L, and the stressed one-day 99% historical VaR. The aggregate P&L numbers we get and the figures the report quotes are in Table 3.

The equity numbers line up well. We get $\pm 280,000$ for an aggregate $\pm 40\%$ shock against their $\approx 300,000$, and the gap is accounted for by how each side aggregates: they appear to compute the portfolio log return as $\sum_i w_i \ln(1 + s)$, which gives the figures in their Table 19 (+12.76%, −15.62%, +23.55%, −35.76%) exactly, while we sum simple P&L euro amounts. The two are economically equivalent for small shocks and diverge a little at $\pm 40\%$, which is what we see.

Risk factor	Shock	Our P&L	Their P&L
Equities (all)	+40%	+ €280,000	$\approx$ + €300,000
Equities (all)	−40%	− €280,000	$\approx$ − €300,000
Equities (all)	+20%	+ €140,000	$\approx$ + €150,000
Equities (all)	−20%	− €140,000	$\approx$ − €150,000
EUR/USD	$\pm 20\%$	$\pm$ €20,000	$\approx \pm$ €22,000
EUR/USD	$\pm 10\%$	$\pm$ €10,000	$\approx \pm$ €11,000
Rate (parallel)	± 2 bp	$\pm$ €40	$\approx \pm$ €10–15
Rate (parallel)	± 3 bp	$\pm$ €60	$\approx \pm$ €10–15

Table 3: Immediate P&L on the EUR 1,000,000 portfolio under the deterministic stress scenarios. “Our P&L” is from our re-implementation; “Their P&L” is read off the report (which only reports orders of magnitude).

The FX numbers are close but not identical: $\pm 20,000$ vs their $\approx 22,000$ for a $\pm 20\%$ shock. The most natural explanation is that we shock only the EUR/USD position itself, while they appear to also pass the EUR/USD move through the USD-denominated equities (XOM, SHEL, GSPC) when re-translating into euros. The report does not actually say which of the two it does, so we cannot verify this; the gap is small but it is systematic.

The rate numbers are the cleanest illustration of the loan-construction issue we flagged earlier. We obtain ± 40 to ± 60 on a ± 2 – 3 bp parallel shift; they report ± 10 – 15 . Both are tiny, but they differ by a factor of three to four, and the reason is mechanical: their loan return is $\ln(1 + (r_t^{\text{DTB3}} + \text{spread})/252)$, so a 1 bp change in the yield translates into roughly $1/252 \approx 0.004$ bp on the daily return. Multiplied by a 20% position weight on a EUR 1 M portfolio, that gives euro impacts in the single digits, which is what they report. We applied the shock as a multiplicative bump on the loan notional, which is slightly larger but still negligible relative to equity and FX risk. Both implementations agree on the qualitative conclusion - the loan is essentially immune to rate shocks at this scale - but neither would catch a real interest-rate stress on a floating-rate liability, because the construction itself rules that out.

Calibration: backtesting FHS–EWMA

We re-ran the rolling out-of-sample backtest of the FHS–EWMA model on the cleaned data, using a 252-day estimation window and a 99% confidence level. The yearly violation counts we obtain and the figures the report gives are in Table 4.

Year	Our viol.	Their viol.	expected number of obs.
2006	0	0	0.04
2007	5	4	2.58
2008	2	2	2.54
2009	2	2	2.60
2010	5	2	2.60
2011	3	3	2.59
2012	3	2	2.58
2013	1	2	2.58
Total	21	17	18.10

Table 4: Yearly 99% VaR violation counts and expected violations under rolling FHS–EWMA.

Both implementations produce a violation rate close to the 1% target - $21/1811 = 1.16\%$ on our side and $17/1787 = 0.95\%$ on theirs - and both pass an unconditional-coverage test at the 5% level. So at the level of the headline conclusion - “*FHS-EWMA is well calibrated for VaR*” - their claim survives our replication, and we agree with the qualitative ranking of FHS-EWMA against the other four models.

We do not reproduce their numbers exactly: we count 21 violations against their 17 over the full sample, with the largest single-year gap in 2010 (5 vs. 2) and smaller mismatches in 2007, 2012 and 2013, and we even disagree slightly on the number of valid out-of-sample forecast days (1811 against 1787, roughly one month of trading). We thus conclude to the following points: the model is well calibrated since both numbers land near the 1% target, and we agree that FHS-EWMA outperforms the parametric VCV alternatives. However, the exact violation count is implementation-dependent, and the report does not give a validator enough detail to arrive at the same number. We treat this as a documentation weakness rather than a model one, and the same caveat applies to every FHS-EWMA result downstream of the backtest.

Assumptions, modelling techniques and risk reporting

The peer group makes few modelling assumptions on top of FHS-EWMA, and the ones they make are reasonable. The only real free parameter is the EWMA decay $\lambda = 0.94$, which they take from the previous assignment rather than estimate; this is the standard choice and we have no objection to it. Beyond that the model is essentially non-parametric. Two choices we would have liked them to state - how they initialise the EWMA recursion and whether they demean the returns - are documentation gaps rather than modelling errors, but they do move the numbers, as the backtest gap in the previous section shows. The usual FHS caveats also apply: $\lambda = 0.94$ reacts slowly when volatility jumps, which we can see in their own 2008 cluster of violations, and a 252-day window cannot produce a tail event larger than the worst residual it contains. There is no real convergence question here - the EWMA recursion is a deterministic filter and the quantile is just a sort - and we get the same numbers on every run of our own code. They report the risk figures clearly: percentages throughout, and the plots back up the tables instead of replacing them. The only piece of the reporting we push back on is the undocumented step of setting the loan VaR to zero, which we already discussed.

4 Limitations

Despite the generally sound framework, several limitations reduce the robustness and real-world applicability of the model.

First, the treatment of foreign exchange risk is incomplete. Assets are kept in their original currencies and no base currency is defined for the portfolio. FX risk is instead proxied by including EUR/USD as a separate position, rather than modelling how exchange rate movements affect the valuation of all foreign assets. This leads to a misrepresentation of the true currency exposure and may underestimate portfolio risk.

Second, the dataset covers the period 2006–2013. While this includes a major stress period, the model is calibrated and evaluated on historical market conditions that may no longer be

representative. As a result, the conclusions and model performance may not generalise well to more recent market environments with different volatility regimes, monetary policies, and structural changes.

Third, the modelling of the loan component is overly simplified. The loan is treated as a floating rate instrument without specifying the reset frequency. This effectively removes interest rate risk, as the return remains consistently positive and stable. Consequently, the loan contributes almost no downside risk, leading to artificially low VaR and ES estimates for the overall portfolio. This assumption weakens the validity of the risk measures and understates the portfolio's exposure to interest rate movements.

Finally, these limitations suggest that the model may struggle to accurately capture risk under more realistic or extreme conditions, particularly when multiple risk factors (FX and interest rates) interact. Addressing these issues would improve both the realism and reliability of the model.

5 Validation Result

Based on model validation, we assign the final model validation result as **Yellow**. While the models are accurate in their methodology and implementation, the model does not get a green due to the simplified loan component that does not accurately capture interest rate sensitivity, and the FX component being included separately also limits and downplays the true exchange rate risk of the portfolio. The model should also be retrained on more recent data to be put into use. With a more recent dataset, restructuring the loan component and assigning a base currency to the portfolio to look at interest rate sensitivity, the model can be deployed.